

- C01023 -

- Digital Systems -



Course Introduction

■ Instructor

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- Course materials: **BKeL**



What will you learn in this class?

- This course provides fundamentals of logic design, such as number presentation and codes, Boolean algebra and logic gates, analysis and design of combinational and sequential circuits.

Why do you study this class?

- It is fundamental course for students in computer science and engineering major.



Learning Outcome

- **L.O.1 Describe digital systems, codes systems, Boolean algebra and logic gates**
 - L.O.1.1 – Describe digital systems, codes systems, main memory, secondary storage, I/O devices
 - L.O.1.2 – Describe Boolean algebra and logic gates
- **L.O.2 Analyze and demonstrate the operation of basic combinational and sequential circuits**
 - L.O.2.1 – Analyze and demonstrate the operation of basic combinational circuits
 - L.O.2.2 – Analyze and demonstrate the operation of basic sequential circuits
- **L.O.3 Design and simulate basic combinational and sequential circuits**
 - L.O.3.1 – Design basic combinational circuits
 - L.O.3.2 – Design basic sequential circuits
- **L.O.4 Assemble combinational and sequential circuits**
 - L.O.4.1 – Assemble combinational circuits
 - L.O.4.2 – Assemble sequential circuits



Tentative Contents

- Number presentation and codes
- Boolean algebra and logic gates
- Combinational circuits
- Sequential circuits



Grading Policy

- Grading
 - Lab: 30%
 - Midterm: 20%
 - Final Exam: 50%



<https://www.hackster.io/projects?ref=topnav>

<http://www.electronicshub.org/iot-project-ideas>

Textbooks

- Digital Systems: Principles and Applications (12th Edition) – Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss.
- Internet
- *Fundamentals of Digital Logic* – Stephen Brown, Zvonko Vranesic, McGraw Hill.
- Digital Design: Principles and Practices– John F. Wakerly, Prentice-Hall



QnA

- https://www.youtube.com/watch?v=j6R_Wmsl79E
- <https://www.youtube.com/watch?v=fi5GJBKb3Ww>
- ...

